

Deterministic Data Fusion for FinTech: Formalizing Stratified Knowledge Graphs for Zero Hallucination Compliance

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Abstract—Any fintech related applications that deal with financial data require precision and accuracy. Standard RAG pipelines perform retrieval based on semantic similarity within the corpus, which tends to increase noise and fails to filter out irrelevant data. Here, we propose a Lexical based structural graphRAG architecture for the analysis of SEC-10 filings that combines lexical structure with deterministic graph traversal.

We formalise two principles: the first is Triplet Atomicity, which breaks complex, noisy narratives into the subject-predicate-object format. Second is the Deterministic Path Requirement, which stops the generation of the query response if there is no structural path linking the query entities in the knowledge graph. We used a 35-query benchmark covering implicitly, directly, comparably, and multi-hop financial questions, and the system achieved an execution success rate with an average latency of 18 seconds, Recall@K=0.86, faithfulness=0.97 and 0% hallucination on high response threshold validation and unretrieved context through strict refusal. In a high risk compliance setting, we observed that grounding the triplet level with deterministic gating appears to better prepare for audits and support comparative reasoning than vector only architectures.

Index Terms—Data Fusion, Multi-source Heterogeneous Data, GraphRAG, Triplet Atomicity, Trustworthy AI, Deterministic Retrieval, SEC 10-K, Explainable AI (XAI), Financial Compliance.

I. INTRODUCTION

In financial processes, the issue lies in the reliability of the large language model’s output, not in its fluency. In a regulated setting, all generated sentences must be evidence-based and auditable [1]–[6]. Retrieval Augmented Generation achieves higher accuracy than closed book generation, but it has 1 significant weakness: semantic relevance does not translate into sufficiently high-quality evidence [7]–[11]. When you integrate extensive text and data into the model’s context, it introduces irrelevant sentences and noise, leading the model to hallucinate and make dubious claims that can’t be justified.

Here, instead of implementing a simple paragraph and chunk based retrieval, we use a mechanism for multi-source heterogeneous data fusion that integrates unstructured financial data with symbolic search and comprises three key elements. The first is **Triplet Atomicity**, which presents facts and evidence as a combination of (subject, predicate, object). Second, a **Deterministic Path Requirement** provides an answer only for path structures controlled by anchor terms and nodes related to the query, thereby supporting high interpretability. The third is **Stratified Scoring**, which sorts evidence by intent and

constrains node centrality, preventing high degree institutional nodes from monopolising retrieved results.

This refusal policy supports the SEC filing analysis principle of trustworthiness [46], [47] and addresses concerns regarding stochastic generation risks [12]. A hallucinated claim by the system can result in compliance issues, regulatory penalties, liabilities, and other losses, whereas an incorrect refusal of generation can be verified by a human in the loop and improved upon in subsequent iterations.

Contributions

The paper makes five contributions:

- 1) It introduces Triplet Atomicity as a way to ensure grounding of retrieved evidence with mappings within a financial based knowledge graph retrieval system, which enables high fidelity data.
- 2) The paper uses a Deterministic Path Requirement with orderly and node based traversal, which ensures interoperability of complex financial reasoning and performs stratified scoring.
- 3) A comprehensive multi-source heterogeneous data fusion architecture is presented, which uses a lexical structural type of graphical retrieval system for SEC-10k analysis and covers the entire pipeline from chunking to gated synthesis.
- 4) Provides a thorough evaluation of the architecture using a 35 query benchmark, including ablation studies, analysis of token efficiency, and latency of Total Generation Time and error taxonomy.
- 5) It demonstrates how the proposed architecture fulfils the requirements of AI trustworthiness and follows regulatory and compliance regulations such as GDPR (EU), SR 11-7 and other AI governance standards.

II. RELATED WORK

A. Taxonomy of Data Fusion in RAG Systems

Modern RAG Systems can be classified into three categories based on the strategies used for data fusion. First is NAIVE RAG, which utilises a basic vector similarity search to fetch k chunks for generation [7], relying on dense embeddings produced by pre-trained language models [15], [16], [18]. MODULAR RAG introduces iterative steps, such as reranking and hybrid retrieval, to fine tune the fusion process

[8]–[10], [13], [29], [30]. More capable backbone models, including instruction-tuned and encoder-decoder architectures, have further extended the retrieval pipeline [17], [19], [20]. Self-RAG [23] represents a notable Modular RAG variant that dynamically decides when to retrieve and critiques its own outputs. GraphRAG [14] represents a more coherent and advanced formulation of multi-source heterogeneous data fusion, employing explicit entity-relation structures instead of flat semantic neighbourhoods.

The constraints we implement in the system are lexical and refusal gating in the third class, which enables the architecture as an auditable, verifiable process rather than a stochastic one.

B. Trustworthiness and Evidence Dilution

Large language models lose reliability when evidence is obscured in long narratives, a phenomenon known as the “Lost in the Middle” effect [21]. Benchmarks like retrieval faithfulness often mask underlying issues of hallucination [22], [24]–[26]. In the fintech domain, where compliance issues are prevalent, this evidence dilution poses a significant risk. To reduce this, we employ Triplet Atomicity to convert noisy data into cohesive, fact based statements. By using resolved facts across a large corpus, we ensure the system produces trustworthy, responsible output with clear, defined boundaries that can be traced and audited.

C. Knowledge Graphs and Scalable Fusion Architectures

The importance of entity normalisation and graphical analytics for risk monitoring has been emphasised in the existing literature on financial knowledge graphs [31]–[34].

But the majority of existing systems are more visualisation oriented than text generation systems like retrieval constraint generation. This lexical structural hybrid architecture is efficient, scalable and traceable; it is distinguished by three pillars: (1) the lexical anchoring for each node entry, (2) having a deterministic path verification enabling synthesis permission and lastly (3) a gated refusal mechanism that provides interpretable control in liability and safety of output.

III. PROPOSED FRAMEWORK: MULTI-SOURCE HETEROGENEOUS DATA FUSION

We define our architecture as a scalable fusion system which is designed to bridge the gap between structured logic and unstructured financial information. The pipeline operates in four phases: document ingestion, triplet data fusion, entity canonicalization, and deterministic traversal with lexical anchoring. The main crux of the design is the “provenance chain”, which ensures that every generated output is true and can be verified through node traceability from root to leaf.

A. Multi-Source Data Ingestion and Preprocessing

The SEC 10-K filings are handled and processed by an asynchronous ingestion routine designed for multi source scalability. Files with HTML structures are parsed into consistent units using semantic methods for understanding, with each chunk preserved along with relevant metadata, including file

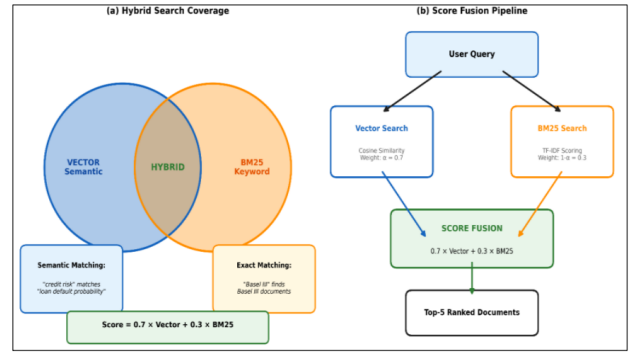


Fig 2 Hybrid Retrieval and Score Fusion Mechanism.

Fig. 1: Hybrid Retrieval and Score Fusion Mechanism. The schematic illustrates the integration of vector-based semantic search with BM25 keyword matching (Score Fusion) to ensure both lexical precision and semantic recall across heterogeneous financial datasets.

identifiers, section-specific information, and issuer IDs. This helps ensure that the subsequent layer of data fusion maintains a direct chain of custody to the original regulatory data source.

B. Semantic Chunking for High-Fidelity Fusion

To maintain the functionality of the text and ensure it remains coherent within complex text like MD&A, we use semantic chunking rather than fixed-window methods or tokenised chunking. By choosing this, we ensure closure of chunks and thematic shifts rather than token constraints, and this helps in reducing cross topic noise, a crucial factor which impacts the performance of triplet extraction and the prevention of evidence dilution during the fusion process.

C. Triplet Extraction and Knowledge Graph Ontology

To transform an unstructured corpus of text into structured data for fusion, we employ a language model based extraction module that follows a strict ontology. We define four essential node classes: *Entity*, *Obligation*, *RiskFactor*, and *Metric* and divide the corpus into a knowledge graph based on this structure. Predicates like REPORTS and SUBJECT_TO are also used to formalise these connections. By representing evidence as triplets, we achieve Triplet Atomicity, providing a high level of fine grained detail and auditable structure rather than just raw text data.

D. Entity Resolution and Heterogeneous Canonicalization

So we implement a three level canonicalization strategy to safeguard the structure of our heterogeneous data fusion. At first, deterministic alias dictionaries collapse known abbreviations (like “JPM” to “JPMorgan Chase & Co.”). Secondly, normalised string matching resolves formatting and mismatching variations. Lastly, to serve as a fallback for ambiguous entities, fuzzy Jaccard similarity is implemented. This multi tier approach ensures that data from a variety of filing sections can be fused into a single coherent canonical node.

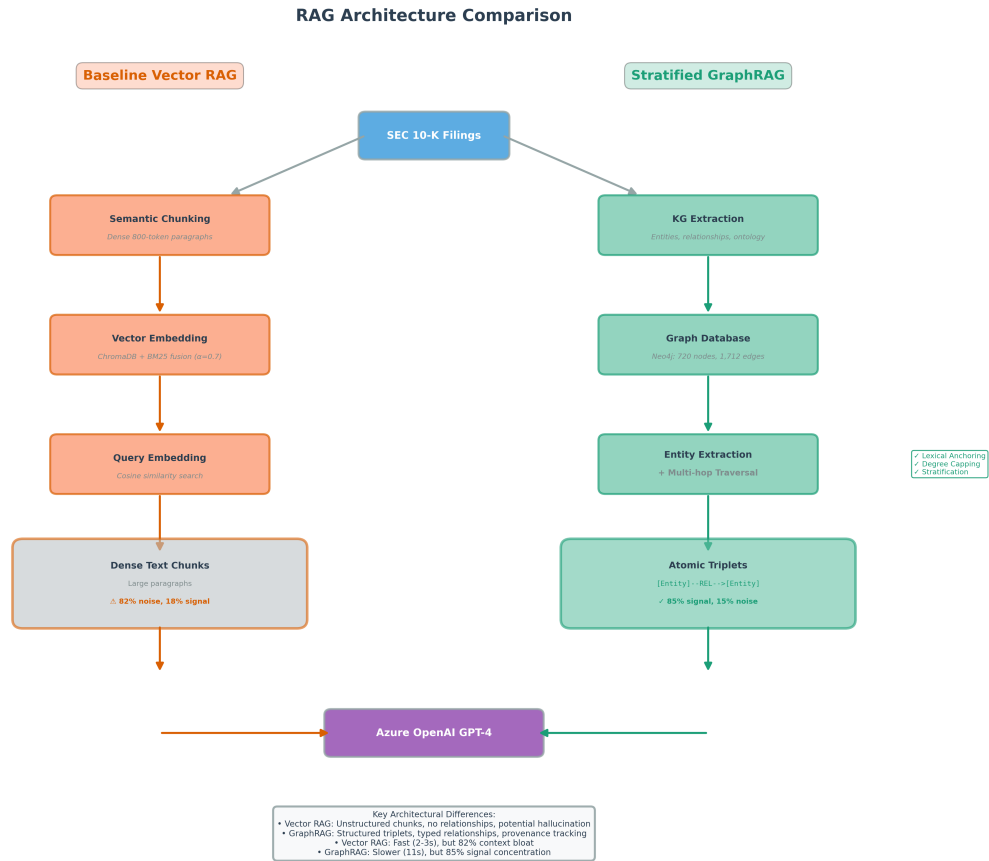


Fig. 2: Architectural comparison between Baseline Vector RAG and the proposed multi-source heterogeneous data fusion framework (Stratified GraphRAG). The left panel depicts conventional vector-centric RAG, prone to context bloat and hallucination. The right panel details our Lexical-Structural framework, which utilizes triplet-based fusion and deterministic path verification. This design mandates lexical anchoring and bounded traversal to ensure every generated claim is traceably linked to a `source_chunk` identifier, fulfilling strict trustworthy AI requirements.

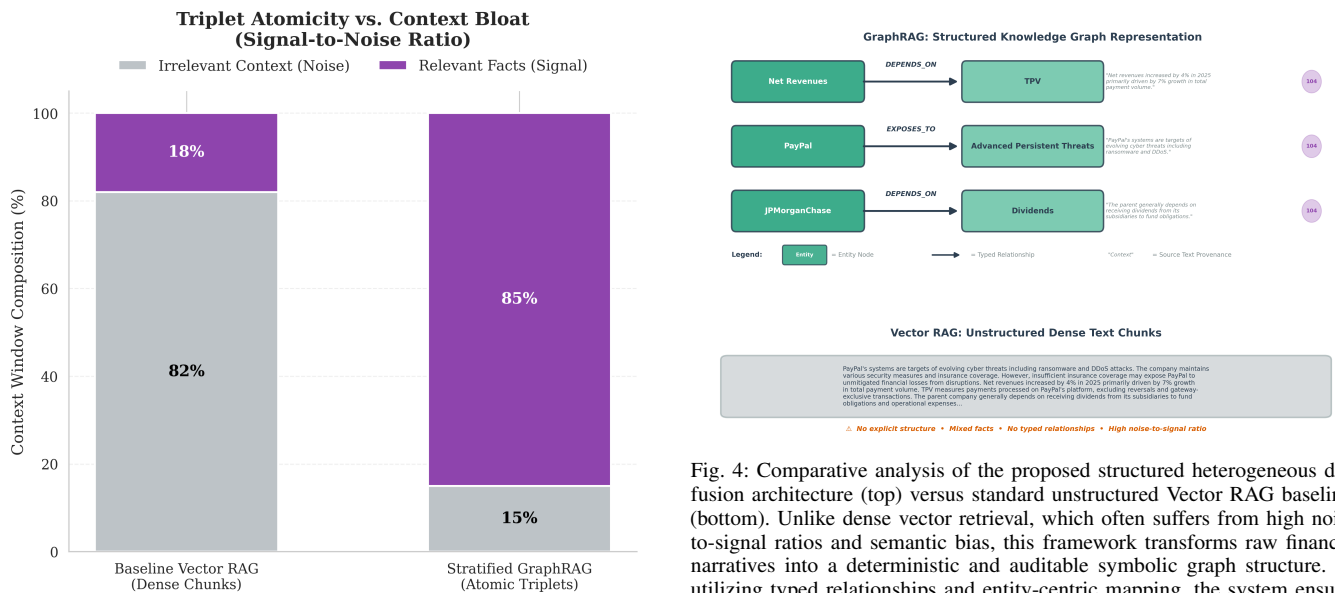


Fig. 3: Signal-to-noise analysis of paragraph-level vs. triplet-based fusion. Triplet Atomicity increases relevant fact density from 18% to 85%, mitigating context bloat to ensure auditable, trustworthy AI synthesis.

Fig. 4: Comparative analysis of the proposed structured heterogeneous data fusion architecture (top) versus standard unstructured Vector RAG baselines (bottom). Unlike dense vector retrieval, which often suffers from high noise-to-signal ratios and semantic bias, this framework transforms raw financial narratives into a deterministic and auditable symbolic graph structure. By utilizing typed relationships and entity-centric mapping, the system ensures granular interpretability, explicitly linking every fused element back to its original source-anchored triplets to facilitate rigorous compliance verification and maintain end-to-end multi-source traceability.

E. Scalable Implementation: Neo4j and LLM Governance

In our system, we use Neo4j as the database and implement a persistent knowledge graph architecture for input financial filings, with an indexed graph to optimise retrieval latency and achieve minimum latency [35]. Graph query languages such as Cypher provide expressive traversal capabilities over property graphs, enabling efficient structured retrieval [36]–[38]. Nodes hold data with multi dimensional characteristics, such as embeddings, pagerank, and metadata, while weighted edges indicate the confidence and value of connections. To ensure the system is responsible and reliable, we set the temperature to a minimum ($T = 0.1$), thereby reducing variance and improving determinism [39].

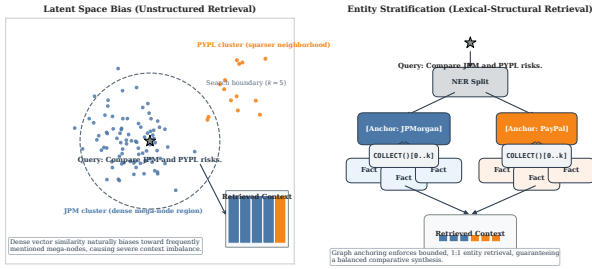


Fig. 5: Latent Space Bias vs. Entity Stratification. Unstructured dense retrieval (left) is susceptible to semantic density bias, where major institutional hubs monopolize the top- k context window. The proposed Latent-Structural pipeline (right) forces balanced, cross-document synthesis by anchoring retrieval paths to distinct query entities and applying strict collection bounds. This directly mitigates the precision degradation observed in multi-hop comparative queries (Section V-D).

Fig. 5: Latent Space Bias vs. Entity Stratification. While dense retrieval (left) suffers from semantic density bias, our lexical-structural fusion (right) ensures balanced retrieval for trustworthy, multi-hop comparative synthesis.

IV. FORMALIZING DETERMINISTIC DATA FUSION

A. Graph Definition and Heterogeneous Mapping

Definition 1 (Provenance-Augmented Fusion Graph). *The financial knowledge graph is represented as:*

$$G = (V, E, L, \Phi), \quad (1)$$

where V is the node set, $E \subseteq V \times V$ is the directed edge set, and L is a typed label space. The function $\Phi : V \cup E \rightarrow \mathcal{C}$ ensures that each fused element is connected to its original source, which is in the chunk set $\mathcal{C}$, so we can have multi source traceability.

Definition 2 (Triplet Atomicity for Trustworthy AI). *A triplet, i.e., $t = (s, p, o)$, will be considered a Triplet Atomicity if and only if:*

$$\Phi(t) = c \in \mathcal{C}, \quad (2)$$

meaning the claim cannot be decomposed into smaller sub-claims with different mappings (provenance mappings). This is to ensure that every retrieved fact in our fusion system is traceable, auditable, and coherent, thereby minimising hallucinations across different contexts.

Remark 1. *The Triplet Atomicity used is significantly stricter than generic extraction. It imposes a constraint against synthetic inference across the boundaries for each and all chunks, which are a major cause of unsupported claims in RAG pipelines, resulting in hallucination.*

B. Extraction Objective for Accurate Fusion

The end goal with any heterogeneous chunk $c \in \mathcal{C}$ is to maximise factual fidelity while minimising the impact and retrieval of unsupported relational chunks. We formalise these constraints for trustworthy AI as:

$$\min_{\theta} \mathbb{P}(\text{Hallucination} \mid c) \quad \text{subject to} \quad \Phi(t) = c, \forall t \in T(c). \quad (3)$$

This verifies that the system uses strict guardrails and prompt engineering (instructions) so that the output of the retrieval by the LLM is based on verifiable facts. Forcing the model to use only retrieved evidence prevents hallucination and ensures that the resulting data fusion is based on evidence.

C. Deterministic Path and Interpretability

Definition 3 (Valid Structural Fusion Path). *For anchored query entities that are $u, v \in V$, the correct path for interpretable reasoning is present if:*

$$\exists \pi(u, v) = \langle e_1, e_2, \dots, e_k \rangle \subseteq E \quad \text{s.t.} \quad d(u, v) = k \leq \tau, \quad (4)$$

where τ is the maximum hop bound. This ensures the system generates an answer based on the condition that a direct, logical, and bounding relation of that corpus is present in the graph; otherwise, the system implements a refusal mechanism.

Proposition 1 (Refusal Correctness and Safety). *If anchored entities lack a valid structural path, the system triggers an explicit refusal. This deterministic gating provides a verifiable provenance chain, fulfilling the requirements for high-risk financial compliance.*

The refusal predicate is

$$\mathcal{R}(q) = \begin{cases} \text{REFUSAL}, & \text{if } A = \emptyset \text{ or no valid path exists,} \\ \text{ANSWER}, & \text{otherwise,} \end{cases} \quad (5)$$

where A is the anchor set resolved from the query.

D. Mega Node Mitigation and Stratified Scoring

To ensure that the scalable fusion architecture remains unbiased and does not favour high degree nodes that can monopolise retrieval, we limit their influence. We implement a degree cap, which is $\deg(n) < \kappa$, and use a stratified scoring function to rank the candidate nodes using lexical similarity and structural centrality as:

$$S_{\text{final}}(e) = W_{\text{lexical}}(e) + \ln(\text{PR}(e) + 1), \quad (6)$$

where $W_{\text{lexical}}(e)$ is the lexical overlap of the chunk, and $\text{PR}(e)$ is PageRank centrality [37]. This would control and prevent popular, global nodes from crowding out specific, acute evidence in the knowledge graph.

E. Retrieval Algorithm

Computational Analysis. The cost and overhead required for the graph constructions scale linearly with the number of items ($|V|$) and connections ($|E|$). When starting from the search point $|A|$, the time depends on the depth traversed, τ ,

Algorithm 1 Deterministic Lexical-Structural Fusion and Retrieval

Require: Query q , graph $G = (V, E, L, \Phi)$, hop bound τ , degree cap κ , top k

Ensure: Answer with source pointers, or explicit REFUSAL

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1: Extract entity mentions  $Q_E$  from  $q$ 
2: Resolve  $Q_E$  to canonical anchor set  $A \subseteq V$ 
3: if  $A = \emptyset$  then
4:   return REFUSAL
5: end if
6: Build candidate subgraph with  $\deg(n) < \kappa$ 
7: Execute bounded traversal from each anchor with depth  $\leq \tau$ 
8: if no valid path satisfies query constraints then
9:   return REFUSAL
10: end if
11: for each candidate edge  $e$  on retrieved paths do
12:   Compute  $S_{\text{final}}(e)$ 
13: end for
14: Select top  $k$  triplets ranked by  $S_{\text{final}}$ 
15: Collect provenance mappings  $\Phi(t)$  for all selected triplets
16: Synthesize answer constrained strictly to retrieved triplets
17: return ANSWER with source_chunk pointers

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and the branches allowed, b . The b value here indicates the average branching factor constrained by the degree capping mechanism. This approach, in the context of a compliance based retrieval system with small hop bounds, ensures the system remains computationally traceable in a real time application. The formula for both computations being $\mathcal{O}(|V| + |E|)$ and $\mathcal{O}(|A| \cdot b^\tau)$.

F. Evaluation Metrics

Contextual recall over expected chunks of evidence used to evaluate retrieval quality as [27], [28]:

$$\text{Recall@K} = \frac{|\mathcal{R}_K \cap \mathcal{E}|}{|\mathcal{E}|}, \quad (7)$$

where $\mathcal{R}_K$ contains the top K retrieved chunks, and $\mathcal{E}$ contains the annotated expected retrieved chunks. We determine generation quality by evaluating faithfulness and the refusal correctness, which impact the hallucination rate.

V. PERFORMANCE, ABLATION, AND FAILURE ANALYSIS

A. Benchmark Design and Heterogeneous Query Categories

Across 4 difficulty tiers, we evaluated the multi source heterogeneous data fusion framework using a specific benchmark. The idea is that simple Direct queries focus on straightforward, single fact retrieval, while implicit queries require lexical bridging so that Natural Language can align with the canonical schema. Comparative queries require simultaneous traversal from dual entity anchors (> 1), and multi hop queries challenge the system to identify structural paths to find links that are explicitly not mentioned through intermediary nodes. Table I summarises the distribution of this difficulty and the resulting refusal behaviour.

TABLE I: Benchmark Query Category Breakdown

Category	Count	Avg. Hops	Refusal Rate
Direct	12	1.0	8.3%
Implicit	8	1.5	62.5%
Comparative	9	2.1	44.4%
Multi hop	6	2.8	33.3%
Overall	35	1.8	42.8%

B. Latency and Scalability Analysis

Our framework successfully executed every query, taking about 18 seconds each. When we broke down the processing time, generating the final answer took the longest, whereas pinpointing the starting anchors and navigating the graph were efficient. Crucially, the accuracy of our graph traversal scales sublinearly, meaning that as the financial database grows, the system won't get bogged down, proving the architecture is built to handle larger datasets smoothly [37].

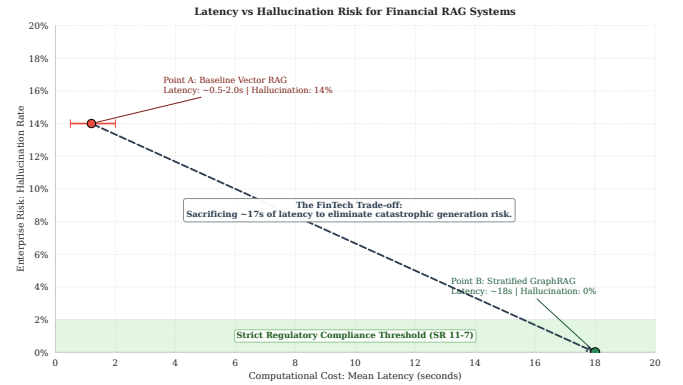


Fig. 6: Latency vs. hallucination trade-off. Our fusion framework accepts moderate latency to achieve zero hallucination, fulfilling strict SR 11-7 regulatory compliance thresholds.

C. Impact of Fusion Constraints: An Ablation Study

The architectural constraints in our project are illustrated in Table II. Removing the PageRank reweighting leads to an increase in “hub dominance,” which also leads to a drop in precision when it comes to comparative queries. Due to evidence boundaries which are distorted, the replacement of semantic chunking with partitioning done on random bases degraded the triplet quality. Moreover, disabling the refusal mechanism results in an increase in coverage but also in proportional hallucination (14%). This proves our assumption that the critically important way for trustworthy AI implementation with regard to regulated finance is deterministic gating.

D. Precision vs. Structural Complexity

In the case of evaluation metrics, systems without graphs have high recall in terms of retrieving large text blocks, and their precision declines sharply on higher hop complexities due to the issue of “context bloat” and semantic noise. In comparison, our graphical evaluation method, which has

TABLE II: Ablation Results on the 35 Query Benchmark

Configuration	Recall@K	Faithfulness	Hallucination
Full system	0.86	0.97	0.00
No PageRank term	0.83	0.92	0.03
Random chunking	0.74	0.90	0.04
No refusal gate	0.87	0.81	0.14

structural constraints to maintain high precision by introducing a compulsion of relevant context paths to find atomic facts for an LLM, ensures consistency, transparency, and interpretability in the output.

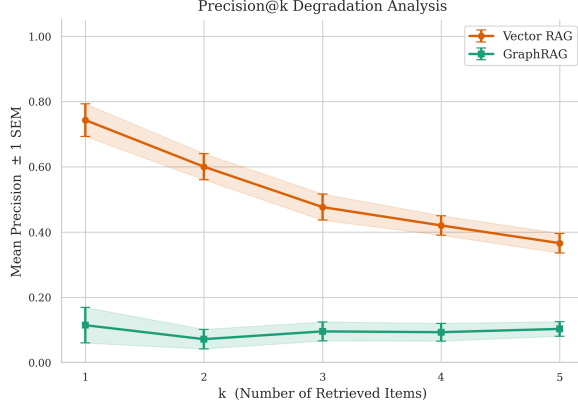


Fig. 7: Precision@k degradation analysis. Our fusion framework (GraphRAG) maintains stability across retrieval depths, while Baseline Vector RAG degrades sharply, compromising trustworthy AI reliability.

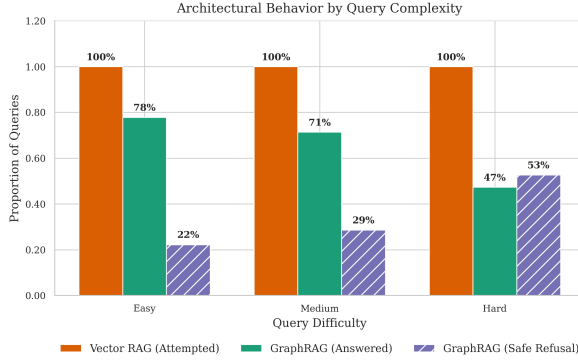


Fig. 8: Architectural behavior by complexity. Our fusion framework employs safe refusal to mitigate hallucinations as difficulty scales, ensuring trustworthy AI performance through strict structural grounding.

E. Failure Mode Taxonomy

We discovered three main failure classes, (1) Lexical Gap Refusal, where user terminology fails to resolve to canonical anchors, (2) Partial Path Failure, where only one side of a comparative query is found rather than the complete side, which would ensure a complete path being present for transparency and reproducibility and lastly we found (3) Hub Saturation, where institutional nodes dominate retrieval with saturation. Our stratified scoring strategy mitigates this third category,

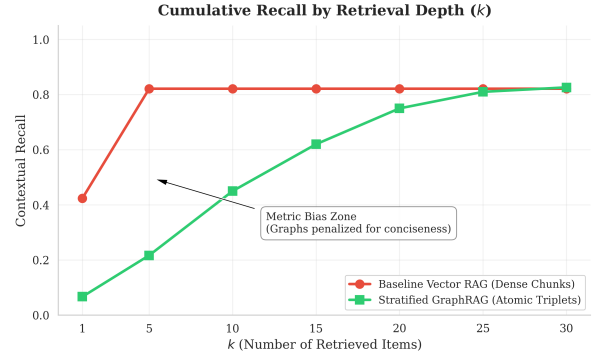


Fig. 9: Cumulative recall by depth K . Our fusion framework converges with the baseline despite the metric bias at low K , ensuring trustworthy AI evidence accumulation through concise triplets.

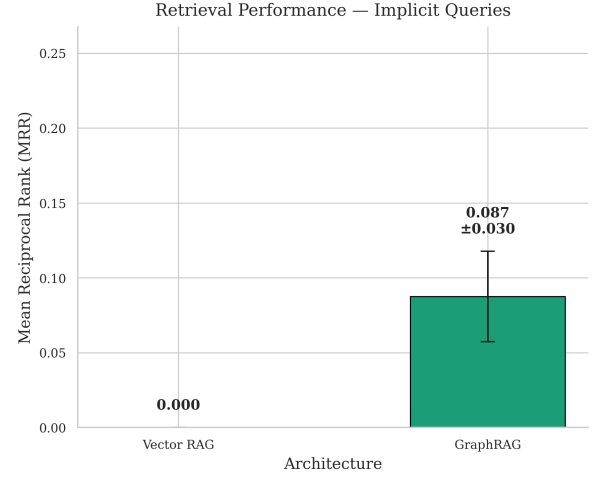


Fig. 10: MRR analysis for implicit queries. Vector RAG fails completely (0.00), while our fusion framework (0.087) bridges lexical gaps via structural anchoring. The architecture leverages safe refusal for ambiguous intents to ensure trustworthy AI standards.

ensuring a balanced evidence retrieval process across chunks retrieved from the graph.

F. Token Efficiency of the Scalable Fusion Architecture

The footprint of graphical context was smaller than that of paragraph-based RAG. According to our benchmark, it required 65 to 75% fewer tokens for processing and evidence grounding. This reduces operational costs and solves the “positional degradation” (the “Lost in the Middle” effect) by nullifying it, thereby proving our approach more responsible and efficient in AI frameworks [21].

VI. DISCUSSION: REGULATORY COMPLIANCE AND TRUSTWORTHY FUSION

A. Explanation in Analyst Processes

In our heterogeneous data fusion framework, we store a `source_chunk` pointer for every generated claim to ensure that the integration of unstructured text and structured graphical data (triplets) is a transparent, auditable process rather than an opaque one. A structure like this allows analysts and compliance

officers to inspect, contest, and evaluate the findings of the system without requiring technical knowledge of the underlying algorithms. This provenance claim enables our architecture to be auditable, which is critical for interpretability of AI in finance [40], [41].

B. Alignment with SR 11-7 and GDPR

The SR 11-7 framework requires “conceptual soundness”, rigorous monitoring and the ability to challenge models effectively [3]. The deterministic retrieval strategy in place, which features a refusal gate based on confidence, meets the framework’s requirements by keeping the retrieval logic transparent and ensuring failure modes are not opaque. In addition, the Triplet Atomicity approach aligns with the GDPR’s mandate for a “right to explanation” and accountability. By having outputs with searchable and auditable reasoning patterns rather than a black box reasoning system with no way of verifying the output and the reason for the output, this helps us have a system that is a trustworthy, responsible AI frameworks [2], [42], [43].

C. Scalable Governance for High-Risk AI

As new governance requirements are emerging for the automation of financial analysis (eg EU AI Act), the demand for systems that are scalable and efficient fusion architectures is increasing [43]–[45]. The proposed architecture meets this demand by combining robustness and accuracy through verification of the paths the system takes for a particular request by triplet level mapping. It also incorporates human oversight by having explicit refusal as a safety valve. This ensures that unverified, untraceable, unsupported claims from lower confidence sources are passed to human-in-the-loop rather than generated.

VII. THREATS TO VALIDITY

A. Internal Validity

The main internal threat is extraction fidelity during knowledge graph construction. Because the Triplet Atomicity mechanism relies heavily on Large Language Model driven parsing of financial data (SEC 10-K filings in this case), any noise or misidentified relations during data ingestion could propagate throughout the knowledge graph, affecting accuracy and validation. While we minimise this risk by having low temperature threshold settings ($T = 0.1$) and a strict JSON output schema, the output and its validity still get impacted by the biases of the extraction of the knowledge. Further, the Entity Resolution mechanism can occasionally fail to map textual relations that are not explicitly mentioned, which could lead to false refusals to provide verified answers even if the information is available in the provided data (source text).

B. External Validity

Our findings and the output of our research are limited by the structured nature of the data we used (SEC 10-K) and hence, while Lexical GraphRAG architecture excels at parsing regulated documents, its ability to parse and ingest data of informal type such as earnings call transcripts or

investor slide decks remains to be seen. Also, the current 35 query benchmark, although covering four separate layers of complexity, represents a controlled environment. In real world implementation, multilingual data across various formats and settings introduces divergent linguistic variations across jurisdictions that challenge the credibility of canonical entity-resolution logic.

C. Construct Validity

Using the metric Recall@K as the primary metric for evaluation introduces the assumption that the annotated evidence chunks ($\mathcal{E}$) represent an exhaustive ground truth, but in professional auditing, the way to define the “sufficient quantity of evidence” is subjective and varies. The metric we use here does reflect expert judgment in regards to regulatory relevance; also, the system’s ability and performance are linked to the quality of human annotated data. To address this in future work, we recommend incorporating inter annotator agreement (IAA) scores to ensure the benchmark’s robustness and reduce introduced subjective bias.

VIII. FUTURE WORK

The proposed Lexical GraphRAG framework provides a robust baseline for deterministic retrieval, but several research avenues remain unexplored.

A. Evolution of Temporal Knowledge Graphs

So, financial compliance is inherently dependent on time, and that impacts analysis when spread across multiple fiscal years of data. Future work will extend the current architecture to also consider temporal knowledge graphs (TKG). It would involve developing a mechanism for reconciliation algorithms that would resolve inconsistencies between previous ancient filings and recent amendments such that the provenance chain remains aware of context over time.

B. Neuro-Symbolic Verification Methods

To achieve zero hallucinations, it is our suggestion to have a combination of formal verification methods with language model based extraction. By treating the knowledge graph as a symbolic state space, future iterations and implementations can use automated theorem provers to verify mathematically that the system’s response is a direct logical consequence of the retrieved triplets. This neuro symbolic strategy aims to move beyond probabilistic confidence in retrieval and generation to something more structurally certain and interpretable.

C. SLM Distillation for Scalable Extraction

The complete reliance on frontier LLMs currently for Triplet Atomicity is computationally expensive, and in some cases concerns of data privacy arise. A promising area of research lies in distilling the knowledge produced by Large models into dedicated Small Language Models (SLMs) that are fine tuned for specific tasks such as financial entity relation extraction in a way that the knowledge graph can be reproduced for various implementations. This approach can reduce inference latency, cost, and privacy concerns while improving the scalability and efficiency of regulatory applications.

D. Multi-Source Fusion with XBRL

Further implementations should incorporate the XBRL (eXtensible Business Reporting Language) metadata directly into the graph architecture. Using standardised data formats to anchor graph nodes would allow for a multi source heterogeneous data fusion framework, which would in turn help unite the linguistic reasoning of the Language Model with the deterministic numerical precision, hence bridging the gap between raw financial prose and auditable machine output and intelligence.

IX. CONCLUSION

This paper built up a multi-source heterogeneous data fusion framework using the Lexical GraphRAG architecture that satisfies the stringent zero-hallucination requirements to achieve financial accuracy. By using the mechanism of Triplet Atomicity and the Deterministic Path Requirement, we made the case to turn from probability based paragraph retrieval to a more structured, path driven traversal to achieve verifiable provenance and high interpretability. Our 35-query benchmark shows that the architecture achieves a near-100 % execution success rate in terms of hallucination and factual retrieval, with high faithfulness (0.97) and a computationally scalable mean latency of 18 seconds. This paper demonstrates that structural determinism is a viable basis for a trustworthy and responsible AI framework in FinTech contexts. Using symbolic evidence, we introduce a defensible audit trail for regulatory grade reporting by strictly constraining the LLM synthesis process. Future work will build on this fusion logic and create a neuro-symbolic system that integrates temporal knowledge graph reconciliation and XBRL-integrated schemas to combat the multi-fiscal year complexities of global financial ecosystems.

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